

Sean A'Hearn

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sean-ahearn.github.io/

PROFILE

A motivated Master of Biomedical Engineering student with a strong interest in developing and translating medical technologies into real-world healthcare solutions. Experienced in engineering design, prototyping, and biosignal analysis, with a focus on medical devices and product development. A collaborative and detail-oriented problem-solver seeking to contribute to multidisciplinary teams.

EDUCATION

Master of Biomedical Engineering (Business Specialisation)

Mar 2024 – Dec 2026

The University of Melbourne, Parkville

- Weighted Average Mark of 76.37
- First Class Honours in Biofluid Mechanics, with other top subjects including Biosignal Processing, Bioinstrumentation, Bioengineering Data Analytics, Biomechanics, Medical Imaging, and Marketing Management for Engineers

Bachelor of Science (Biomedical Engineering Systems)

Mar 2021 – Dec 2023

The University of Melbourne, Parkville

- Weighted Average Mark of 77.95
- First Class Honours in 11 Subjects including Biosystems Design, Stem Cells in Development & Regeneration, Experimental Pathology, and Applied Computation in Bioengineering

Victorian Certificate of Education

Jan 2020 – Dec 2020

Xavier College, Kew

- The Dalton Award for Academic Excellence in VCE
- Biology Class Prize

TECHNICAL SKILLS

- Proficient in MATLAB, Autodesk Fusion, LabVIEW, KiCad, and circuit simulation software
- Practical experience with electronic circuits, components, and fundamental circuit theory
- Certified in Electronics and Soldering by The University of Melbourne's Creator Space
- Laboratory experience with hydrogels, PCR, ELISA, electrophoresis, histology, mechanical testing (compression/tension), and fluid shear testing
- Strong skills in statistical analysis, technical writing, project tracking, and scientific reporting

RELEVANT EXPERIENCE

Biomedical Engineering Capstone Project

Mar 2026 – Nov 2026

The University of Melbourne / Royal Melbourne Hospital, Parkville

- Developing a wearable medical device to objectively assess seated postural control in spinal cord injury patients, in collaboration with clinicians at the Royal Melbourne Hospital

Intern Biomedical Engineer

Jul 2025 – Nov 2025

Murdoch Children's Research Institute (MCRI), Parkville

- Tested, validated, and optimised biosignal sensors for a prototype medical device
- Researched materials and supported data-driven sensor selection and design decisions
- Developed a Signal Quality Index (SQI) algorithm to quantify photoplethysmography (PPG) signal quality

Bioinstrumentation Project

Mar 2024 – Jul 2024

The University of Melbourne, Parkville

- Designed and prototyped a functional analogue hearing aid with amplification and filtering, featuring user-controlled volume and hazardous noise alerts

Biosystems Design Project

Jul 2023 – Dec 2023

The University of Melbourne, Parkville

- Designed and developed a wrist-worn device for monitoring UV exposure and body temperature within a team of five, responsible for circuitry, testing, 3D-printed case design, and signal processing

OTHER EMPLOYMENT

Delicatessen Staff Member

Dec 2020 – present

Boccaccio Cellars IGA, Balwyn

- Recognised by managers and customers for fast, friendly, and quality service
- Trained and mentored 15+ new staff members
- Demonstrated strong leadership and verbal communication skills

REFEREES

Associate Professor Jonathan Mynard

MCRI Team Leader / Senior Research Fellow (Cardiovascular Bioengineering)

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